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Refinery Reliability Reinvented: An In-House System for Process Anomaly Detection

Phanishwar Kumar, Hindustan Petroleum Corporation Limited, Mumbai, Maharashtra, India

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Abstract

Process reliability is the foundation of safe and profitable operations in petroleum refining. However, traditional anomaly detection frameworks—built on static alarm limits and manual surveillance—struggle to capture subtle, multivariate deviations in real-time process data. As refining operations evolve to become more dynamic and complex, there is an increasing need for adaptive, data-driven systems that can autonomously identify anomalies, explain their causes, and help mitigate them before they impact operations.

This paper presents the end-to-end design, development, and deployment of an industrial-scale Process Anomaly Detection System (PADS), developed entirely in-house by Hindustan Petroleum Corporation Limited (HPCL). The system's objective is to enhance plant reliability and operational discipline by proactively detecting anomalous process behavior and instrumentation faults using unsupervised learning techniques. Unlike conventional systems that depend on labeled failure data or fixed thresholds, PADS dynamically learns multivariate relationships among process parameters to determine what constitutes normal operation—continuously updating its internal reference states as the plant evolves.

At its core, PADS employs a hybrid suite of unsupervised statistical learning techniques including Moving Window Principal Component Analysis (MWPCA), Hierarchical PCA (HPCA), and Slow Feature Analysis (SFA). MWPCA enables adaptive modeling in non-stationary environments by recalibrating reference baselines over sliding windows. HPCA introduces a physical hierarchy into the anomaly detection framework, facilitating isolation of unit-level or inter-unit issues. SFA detects slow-developing degradation trends such as catalyst deactivation, fouling, or instrument drift, often missed by conventional alarms. Once a deviation is detected through Q^2 and T^2 metrics, the system uses Reconstruction-Based Contribution (RBC) analysis to pinpoint contributing variables and offer root cause visibility.

The system architecture comprises real-time integration with the refinery's Distributed Control System (DCS), a Python-based backend for inference and data processing, and a Django-powered industrial-grade web dashboard for intuitive visualization and operator interaction. This complete solution was built using open-source tools and deployed across more than 20 process units in HPCL's Mumbai and Visakhapatnam refineries. Field application of PADS has demonstrated several operational gains, including early detection of process deviations, valve instability, and sensor drift. Additionally, operators have reported increased

trust in digital diagnostics owing to the transparency and interpretability provided by contribution plots and anomaly evolution timelines.

PADS represents not just a software solution but a fundamental shift in how refineries approach anomaly detection. It replaces reactive alarm-handling with proactive, explainable insights, enabling a structured reliability framework at scale. Moreover, the in-house development allowed HPCL to tailor every aspect of the system—model tuning, hierarchy definition, frontend workflows—to the refinery's specific operating context. This paper highlights how unsupervised learning, when embedded into the daily fabric of plant operations and developed by domain experts themselves, can significantly elevate operational reliability and advance refinery digitization without dependency on external vendors or costly proprietary tools.

By bridging the gap between data science and field operations, PADS sets a precedent for refinery-wide anomaly detection systems that are interpretable, scalable, and resilient to real-world process variability.

Introduction

In the highly integrated and asset-intensive environment of petroleum refining, operational reliability is not merely a metric—it is a business imperative. Every unit shutdown, quality deviation, or safety incident carries tangible consequences in terms of production loss, maintenance expenditure, environmental penalties, and personnel risk. To sustain profitability while ensuring compliance and plant safety, modern refineries must adopt systems that can continuously and accurately monitor the health of both processes and instrumentation.

Despite significant advances in instrumentation and control systems, the prevalent approach to anomaly detection in many industrial facilities still rests on a foundation of static alarm limits, operator intuition, and retrospective analysis. These methods, although serviceable under steady-state or well-understood operating regimes, have increasingly proven inadequate in dynamic, multivariate process environments. Industrial process data is inherently complex—characterized by non-linear interactions, feedback dependencies, and transient operating conditions such as startups, grade transitions, and turndowns. In such environments, simple thresholds cannot reliably distinguish between benign process variability and early signs of trouble.

One of the most visible symptoms of this limitation is alarm flooding, wherein a minor anomaly or transient fluctuation triggers a cascade of alarms across interconnected variables. This phenomenon not only overwhelms operators but also reduces their ability to recognize genuine issues in real time, often resulting in missed early warnings or delayed interventions. Furthermore, when root causes are obscured by secondary effects, operators are left to rely on experience-based troubleshooting rather than data-driven diagnostics.

Recognizing these challenges, Hindustan Petroleum Corporation Limited (HPCL) embarked on the development of an in-house Process Anomaly Detection System (PADS) to serve as a central nervous system for early warning and reliability analytics. The primary goal of PADS is to enable refinery units to transition from reactive alarm-handling to proactive anomaly detection and transparent root cause analysis. To achieve this, the system employs a combination of unsupervised multivariate statistical learning techniques capable of learning the natural behavior of the process without the need for labeled fault data.

Unlike rule-based expert systems or supervised machine learning models that depend on historical failure labels—which are often unavailable, incomplete, or unit-specific—PADS builds a dynamic understanding of "normal" process behavior using only live data streams. It then continuously compares incoming plant data to this learned reference state to identify deviations. More importantly, once a deviation is detected, the system attributes the anomaly to the most influential process variables using statistical contribution analysis, thereby allowing field teams to take targeted and timely corrective actions.

What sets PADS apart is not only its algorithmic core, but also the philosophy behind its development. The system was conceptualized, designed, and implemented entirely in-house by a multidisciplinary team comprising process engineers, data scientists, and software developers at HPCL. This approach ensured deep

domain integration, faster iteration, and alignment with operational realities. It also provided HPCL with complete control over the source code, deployment architecture, and customization pathways—advantages that are often restricted in vendor-supplied or black-box anomaly detection solutions.

Since its inception, PADS has been deployed across more than 20 operating units at HPCL's Mumbai and Visakhapatnam refineries, covering a diverse range of process configurations—from catalytic reforming and hydrotreating to stabilizers and distillation towers. Each deployment was preceded by a detailed process understanding exercise, allowing the modeling strategy to be tuned for unit-specific dynamics.

Beyond the technology itself, the introduction of PADS has had a measurable cultural impact—operators have expressed higher confidence in digital diagnostics due to the system's transparency and interpretability, while engineering teams have begun integrating anomaly insights into maintenance planning and instrumentation health monitoring. The result is not just improved detection but improved response, transforming the reliability paradigm from "alert and react" to "understand and act."

This paper presents the full journey of PADS—from conceptual framework and algorithm design to deployment architecture, field use cases, and observed benefits. It is intended to demonstrate how unsupervised learning, when grounded in real operational knowledge and developed collaboratively by end-users, can unlock new levels of process reliability in modern refining environments.

System Architecture and Methodology

The architecture of PADS encompasses three key dimensions:

1. Algorithmic framework for unsupervised anomaly detection
2. Real-time data ingestion and inference pipeline, and
3. Interpretability mechanisms for root cause analysis.

The application may be divided in to the following parts:

- **Data Integration Pipeline:** Real-time data from the DCS is streamed through secure connectors, preprocessed using filtering, normalization, and then passed to the anomaly detection models. This ensures clean and synchronized multivariate datasets.
- **Modeling Strategy:** The core algorithms—MWPCA, HPCA, and SFA—are orchestrated to detect anomalies across different temporal scales and process hierarchies. Each block outputs T^2 , Q^2 , or drift trajectories, which are aggregated into a composite health index. Upon Anomaly Detection they are passed through domain based thresholds, which are unit specific and categorized into 3 types – Erratic, Stuck and Fast Moving.
- **Visualization and Interaction:** A Django-based web dashboard provides operators with anomaly timelines, contribution plots, and drill-down views. Visualization is designed for clarity in the control room, with operator acknowledgment and feedback tightly integrated into anomaly records.
- **Feedback and Adaptation:** Operator feedback directly influences threshold refinement and variable prioritization, ensuring continuous improvement of model accuracy.

Unsupervised Learning for Anomaly Detection

The Process Anomaly Detection System (PADS) is architected around a suite of unsupervised multivariate statistical learning algorithms—specifically, Moving Window Principal Component Analysis (MWPCA), Hierarchical PCA (HPCA), and Slow Feature Analysis (SFA). Each of these techniques addresses a different class of anomaly: short-term deviations, hierarchical or interdependent disturbances, and long-term degradation patterns, respectively. Together, they form a complementary toolkit that enables PADS to robustly model refinery processes without requiring labeled datasets or predefined fault templates.

Moving Window Principal Component Analysis (MWPCA)

Principal Component Analysis (PCA) is a widely used dimensionality reduction technique that transforms a multivariate dataset into a set of linearly uncorrelated variables called principal components (PCs). These components capture the directions of maximum variance in the data and help in eliminating redundant correlations among process variables.

The classical PCA model is built as follows:

Given a data matrix $\mathbf{X}$ of size $n \times p$ (where n is the number of samples and p is the number of variables), the matrix is centred (and optionally scaled), and decomposed using singular value decomposition (SVD):

$$\mathbf{X} = \mathbf{TP}^T + \mathbf{E}$$

Where:

- $\mathbf{T}$ = Projection of data on principal components
- $\mathbf{P}$ = Eigenvectors of covariance matrix
- $\mathbf{E}$ = Reconstruction error

The principal components retain the structure of the data in a lower-dimensional space, typically capturing 80-95% of the variance in the first few components. This allows the system to reduce noise and focus on systemic variations.

Moving Window PCA (MWPCA) enhances this technique by applying PCA over a rolling time window (e.g., 6-hour or 12-hour blocks). This is crucial in refinery operations where the process mean and variance are not constant over time due to factors such as feedstock changes, ambient shifts, or seasonal patterns. MWPCA continuously updates its reference state in a moving window fashion, enabling localized sensitivity to process drifts.

Two key statistics are monitored for anomaly detection:

- **Hotelling's T^2 (Squared Mahalanobis Distance):**

$$T^2 = \mathbf{t}^T \mathbf{\Lambda}^{-1} \mathbf{t}$$

Where $\mathbf{t}$ is the score vector and $\mathbf{\Lambda}$ is the diagonal matrix of retained eigenvalues. T^2 captures how far a sample lies in the principal component space.

- **Q^2 (Squared Prediction Error):**

$$Q^2 = \|\mathbf{e}\|^2 = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$$

Q^2 reflects how much information is lost in reconstructing the sample, i.e., how far it lies from the PCA model subspace.

If either of these statistics exceeds dynamic control limits—calculated from historical windows using non-parametric methods such as kernel density estimation—an anomaly flag is raised.

Hierarchical Principal Component Analysis (HPCA)

In real refinery environments, process tags are rarely independent. Units are organized in physical and logical hierarchies—e.g., a stabilizer column depends on upstream preheaters, and reactor pressures impact downstream separation. A flat PCA model over all tags may dilute these dependencies and obscure localized issues.

Hierarchical PCA (HPCA) solves this by segmenting the process into logical blocks (e.g., feed preheat, reactor, fractionator, product header) and building separate PCA models for each block. Each local PCA model monitors intra-unit anomalies independently, while their results are aggregated to form a global health index.

This structured decomposition offers several advantages:

- **Improved interpretability:** Anomalies can be traced to specific zones, making diagnosis easier for operators and maintenance teams.
- **Fault localization:** Upstream vs. downstream issues can be distinguished.
- **Scalability:** Models are modular and easier to update during plant modifications.

Mathematically, this involves creating independent score spaces ($T_1, T_2, \dots, T_k$) and monitoring their respective Q^2 and T^2 metrics. In practice, block-wise correlation matrices and loadings are stored, and the outputs are fused in the dashboard visualization layer.

Slow Feature Analysis (SFA)

Whereas MWPCA and HPCA are geared towards detecting rapid or mid-scale anomalies, certain failure modes evolve slowly over time. Examples include catalyst degradation, exchanger fouling, thermal fatigue, or instrument drift. Such patterns often evade standard monitoring techniques, as their signals are masked by normal variability in short-term data.

Slow Feature Analysis (SFA) is a technique originally developed for detecting slowly changing signals embedded in rapidly fluctuating time series. It extracts latent variables whose temporal derivatives are minimal.

Slow features retain high information content but evolve gradually over time. These are ideal for long-term health monitoring.

In PADS, SFA is applied to historical datasets to identify variables with emerging drifts. The resulting slow features are then tracked in near-real-time, and significant deviations from historical envelopes are treated as potential degradation signals.

Anomaly Scoring and Thresholding

Anomaly detection in PADS is driven by the real-time evaluation of T^2 and Q^2 statistics from MWPCA/HPCA models and slow feature trajectories from SFA. However, one of the critical challenges in deploying such systems in live industrial settings is setting robust thresholds that avoid false alarms without compromising sensitivity.

To address this, PADS employs dynamic thresholding based on historical kernel density estimation (KDE) and User Feedback System. For each statistic:

- A rolling history is maintained to estimate the empirical probability density function (PDF).
- Thresholds are set based on confidence intervals (typically 95th or 99th percentile) from the KDE.
- These limits adapt as the process evolves, unlike fixed thresholds that may become obsolete.
- Furthermore, these thresholds are further tuned by the User Feedback System (refer next sections)

Each anomaly instance is logged with:

- Timestamp
- Unit/block identifier
- Contributing variables (from RBC, see next section)
- Anomaly type (Erratic, Stuck and Fast Moving)
- User acknowledgment and feedback (if any)

This structured anomaly record serves as both an operational insight and a dataset for further reliability analysis or model refinement.

Root Cause Diagnosis via Reconstruction-Based Contribution (RBC)

Once an anomaly is confirmed, the next logical step is to diagnose its root cause. PADS leverages Reconstruction-Based Contribution (RBC) analysis, a mathematical method that identifies which variables contribute most to the deviation.

In RBC, the reconstruction error of each tag (difference between actual and reconstructed value from PCA space) is computed. Tags with the highest normalized error are highlighted as potential culprits. This mechanism drastically reduces diagnostic time and enhances operator confidence.

System Deployment and Results

The deployment of PADS is characterized by its real-time integration, secure architecture, and fully in-house development using Python and Django.

Real-Time Monitoring and Integration

The system is tightly integrated with DCS (Distributed Control System) infrastructure to ingest real-time process data. A backend processing engine performs continuous model inference, while a frontend dashboard provides operators and engineers with intuitive visualizations of anomalies detected, contribution plots, and historical reports.

The entire stack—algorithms, backend architecture, and frontend interface—was developed in-house, ensuring seamless integration with HPCL's IT landscape and complete control over customization.

Front-End Development

The entire web interface was developed using Python's Django framework and deployed on HPCL's internal servers. The dashboard presents:

- Live anomaly scores with unit-wise filtering
- Historical trends with zoom, pan, and overlay capabilities
- Contribution plots with variable ranking
- Feedback mechanism for operators to accept, reject, or comment on anomaly detection

The UI was designed in consultation with stakeholders to ensure clarity, usability, and responsiveness on existing workstation setups.

User Feedback and Model Tuning

PADS incorporates a built-in feedback mechanism that allows users to validate or reject detected anomalies directly from the dashboard. This feedback is logged and used to iteratively tune the system's detection thresholds and variable weightings, helping reduce false positives and improve model sensitivity over time. By combining statistical first-principle-based modeling with user experience-driven adjustments, PADS evolves into a more accurate and operator-aligned system. This human-in-the-loop tuning not only enhances detection performance but also fosters trust and continuous engagement from field teams.

Observed Benefits Post-Deployment

The Process Anomaly Detection System (PADS) has been successfully deployed across more than 20 critical process units spanning two of HPCL's major refineries—Mumbai and Visakhapatnam. These deployments include diverse configurations such as continuous catalytic reformers (CCR), naphtha hydrotreaters (NHT), diesel hydrotreaters (DHT, vacuum distillation units (VDU), and Sulfur Recovery units, each with unique process dynamics and control philosophies. Despite the inherent differences in operating conditions,

equipment profiles, and historical performance, the outcomes of PADS deployment have consistently demonstrated tangible operational value.

One of the most valuable impacts of PADS has been its ability to detect abnormal conditions much earlier than conventional systems. In multiple instances, the system has pre-emptively flagged:

1. Early Detection of Hidden Anomalies

- Equipment malfunctions, such as stuck control valves, sudden increased rate of increase in feed surge drum levels, well before temperature, level or pressure limits were breached.
- Sensor and transmitter drift, which would otherwise go unnoticed due to slow progression or operator desensitization. In one case, a thermocouple drift in a reactor bed was identified via increasing contribution scores even though the process appeared to be within limits.
- Sudden process condition changes, such as composition swings during feed transitions or disturbances due to upstream unit instability, allowing operators to take anticipatory actions.
- Off-spec product trends, particularly in stabilizers and reformers, where gradual deviation from quality norms was predicted hours before laboratory samples confirmed the issue.

2. Reduction in Spurious Alarms and Operator Fatigue

One of the longstanding pain points in refinery control rooms is alarm flooding—wherein a single disturbance may trigger a cascade of high and low alarms across interdependent process tags. Such events not only create cognitive overload for control room operators but also reduce their sensitivity to genuine alarms.

- Post-PADS deployment, Operator has a tool to declutter this alarm flood and directly focus on the root cause.
- Detection of Tags that can lead to spurious Alarm if not attended on time.

3. Strengthened Asset Monitoring and Maintenance Culture

Beyond short-term anomaly detection, PADS has begun to influence the long-term approach to asset care. By providing continuous feedback on instrumentation reliability and process drift, the system has encouraged:

- Data-driven maintenance, where field inspection and servicing are prioritized based on PADS-flagged issues rather than time-based routines.
- Cross-functional RCA reviews
- Use of anomaly logs and trend evolution for post-event analysis, helping to identify contributing factors and improve standard operating procedures (SOPs).

In some cases, maintenance tasks previously considered reactive—such as instrument recalibration or control valve tuning—are now being performed proactively based on persistent low-level alerts from PADS.

Discussion

Adaptability in Dynamic Process Environments

Refinery operations often undergo transitions in feedstock, throughput, and environmental conditions. Static anomaly detection frameworks quickly become obsolete in such environments. The window-based PCA and SFA techniques in PADS allow the system to adapt and remain relevant throughout the lifecycle of a unit.

Role of In-House Development

By developing PADS internally, HPCL was able to incorporate deep domain knowledge into system design. Unlike off-the-shelf anomaly detection tools, PADS is tailored to the nuances of refinery operations and instrumentation behaviors.

Conclusion

Summary

The Process Anomaly Detection System (PADS) stands as a proven example of how unsupervised learning techniques—when thoughtfully integrated with domain knowledge and developed in-house—can lead to scalable, interpretable, and high-impact solutions for refinery reliability enhancement. In an industrial context where real-time process data is abundant but labeled fault data is scarce, PADS successfully bridges the gap between statistical modeling and practical operations.

Unlike conventional alarm-based systems or opaque black-box analytics, PADS introduces a structured, data-driven approach to anomaly detection and root cause analysis. The system's architecture is both technically rigorous and operationally grounded, leveraging:

- **Moving Window PCA (MWPCA)** to account for the dynamic nature of real-world processes.
- **Hierarchical PCA (HPCA)** to reflect the physical and logical structure of process plants, enabling fault localization.
- **Slow Feature Analysis (SFA)** to uncover long-term degradation patterns that are otherwise invisible in traditional systems.
- **Reconstruction-Based Contribution (RBC)** techniques for interpretable diagnostics, helping engineers pinpoint variables driving abnormal behaviour.

These capabilities, coupled with an intuitive frontend developed using Django and real-time DCS integration, allow PADS to operate as a plug-and-play digital reliability layer. The successful deployment of the system across more than 20 refinery units has demonstrated clear operational benefits—early anomaly detection, reduced spurious alarms, proactive instrumentation monitoring, and improved user engagement.

Perhaps equally important is the cultural impact. By demystifying statistical analytics and embedding transparency into the system's design, PADS has helped shift operator mindsets from reactive problem-solving to proactive reliability thinking. This transformation is crucial for the long-term success of any digital solution in a high-stakes industrial environment.

Limitations and Challenges

While PADS has shown strong results in HPCL refineries, some challenges remain. Retraining overhead—particularly in data-heavy units—requires careful scheduling during low-load periods. Operator adoption, though improving, initially faced resistance due to perceived complexity; sustained training and interpretability have helped address this. Scalability beyond HPCL will require adaptation to different historian systems, tag naming conventions, and IT infrastructure. These limitations are natural in first-generation deployments but also define the roadmap for broader industrial adoption.

Future Work

Future iterations of PADS will include integration with maintenance management systems and predictive analytics for failure forecasting. Expanding the algorithm suite to include autoencoders and hybrid ML-physics models is also under consideration.

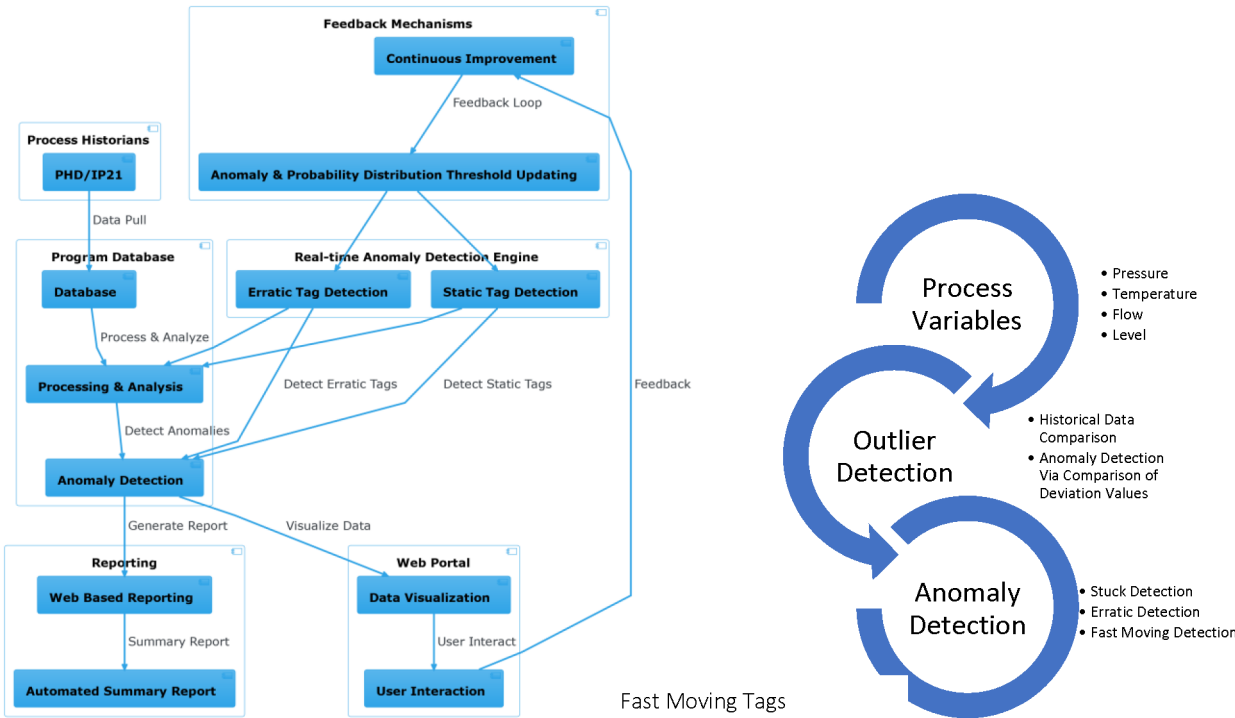


Figure 1—PADS Architecture and Workflow. The figure illustrates the end-to-end workflow of the anomaly detection system, from real-time DCS data ingestion to model inference, anomaly scoring, and visualization dashboard.

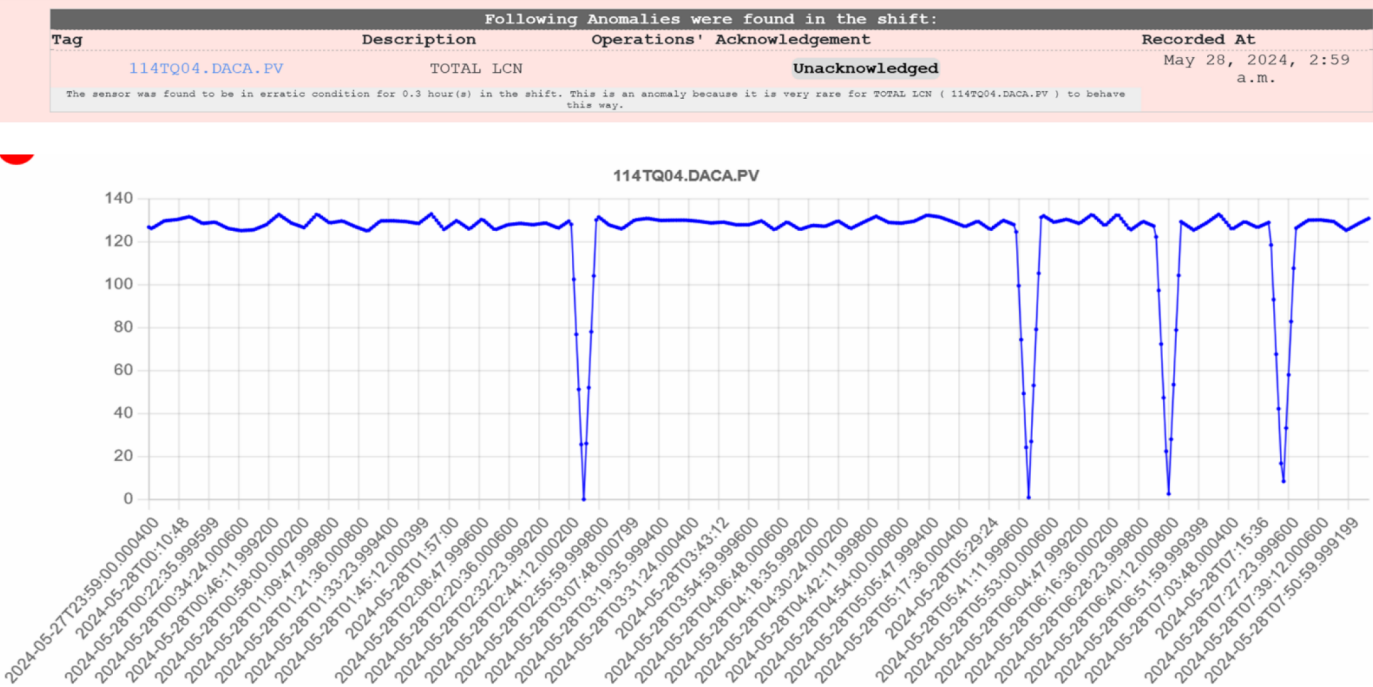


Figure 2—Actual Dashboard and Feedback System. The figure illustrates the actual PADS Dashboard and Operator Feedback System

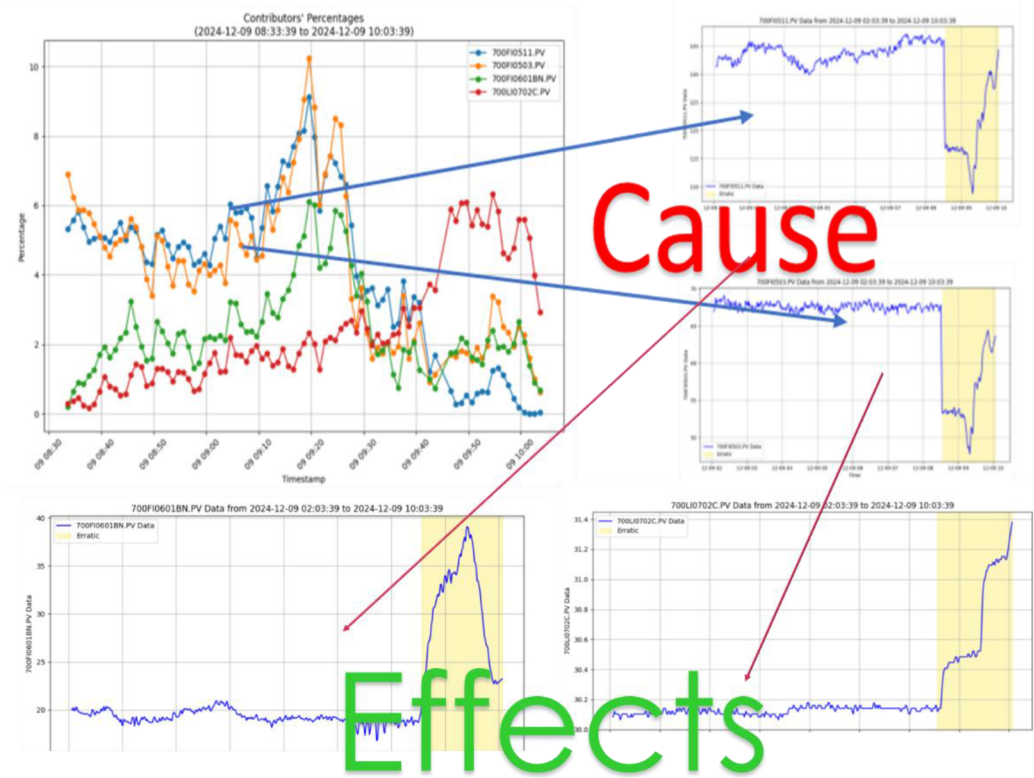


Figure 3—RBC based Root Cause Analysis.

Table 1—Overview of Algorithms and Functions in PADS. This table summarizes the unsupervised algorithms employed and their specific roles in anomaly detection and diagnostics.

Algorithm	Functionality	Purpose
MWPCA	Dynamic PCA in Moving Windows	Adapt to non-stationary process states
HPCA	Hierarchical Modeling	Capture inter-unit and intra-unit dependencies
SFA	Trend Detection	Identify slow-changing faults or drifts
RBC	Root Cause Analysis	Attribute deviations to specific tags or sensors